

Application of Noise Analysis to Safety-Related
Diagnostics and Assessments*

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FORT ST. VRAIN

We recently used noise analysis methods to aid the NRC in assessing in-core temperature fluctuations (presumably due to graphite core block movements) at the Fort St. Vrain (FSV) gas-cooled reactor. We compared ex-core neutron noise signals and pressure vessel displacement probe signals to infer the predominant direction of movement of the graphite core blocks in the FSV. Noise techniques were also used to confirm that the movements of individual blocks were more or less random in nature (i.e., the total core did not move instantaneously as a unit) and that the movement did not result in significant total core reactivity changes.

THREE MILE ISLAND

Oak Ridge National Laboratory (ORNL) and its consultants from Science Applications, Inc., and Technology for Energy Corp. used noise analysis methodologies at TMI-2 to provide: (1) assessment of the size of a bubble of noncondensable gas in the primary system and assessment of the structural integrity of the reactor vessel internals by using the loose-parts monitoring system; (2) monitoring for incipient failure of critical post-accident instrumentation (pressurizer level indicators) and for bulk coolant boiling in the reactor core; and (3) confirmation and operator

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assistance in degasification of the primary system prior to establishment of natural circulation cooling.¹

BASELINE SIGNATURES

In addition to the preceding applications of noise analysis to current safety-related problems, we are continuing to broaden the NRC noise analysis capabilities by acquiring baseline noise signatures for full-power plant operation and by evaluating new methods for understanding and applying noise signals for safety assessment. Our previous experience in the application of noise analysis for diagnosing and assessing abnormal plant behavior had shown a need for baseline, or reference, signatures for normal plant behavior against which to compare abnormal signatures. This need was again emphasized at TMI-2, where we were asked to monitor noncondensable gas in the primary water, detect boiling, detect loose parts, etc., with the reactor in a hot shutdown condition and with only one primary coolant pump in operation. Noise analysts could have performed these tasks much better if baseline signatures for pressure, temperature, and vibration signals had been readily available.

Recognizing this need, the NRC had asked ORNL even before the TMI-2 accident to acquire limited neutron noise baseline data from a few PWRs (we acquired signatures from two PWRs this year). However, based on subsequent TMI-2 experience, we are strongly recommending that baseline signature acquisition be expanded to include signals other than neutron noise and to include some measurements at other than full-power operation.

In connection with these studies, we plan to install a continuous monitoring system in a plant to automatically acquire and catalog noise signatures for a number of selected signals; then it will detect any deviations from baseline conditions that might appear. We also hope to eventually "train" the system to perform some diagnostic tasks (possibly detect PWR core boiling, loose core barrel, or abnormal fuel element vibration).

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ANALYTICAL PREDICTION OF NOISE SIGNATURES

We have also developed and performed an initial evaluation of a stochastic modeling methodology that promises to be a great asset in reactor noise analysis.^{2,3} This technique will allow noise analysts to predict by analytical methods the sensitivity of noise analysis for detecting postulated anomalies. For example, we plan to use this approach to calculate the minimum amount of in-core coolant boiling that can be detected by ex-core neutron detectors in a pressurized-water reactor.

LOOSE-PARTS MONITORING

In another effort, we assessed methods currently used by the nuclear industry to detect, locate, and determine the size of loose parts in U.S. commercial power reactors.⁴ We further assessed the methodology through fundamental research and measurements, including studies on the pressure vessel of the Experimental Gas-Cooled Reactor at ORNL. Important outcomes of these tests have been recommended procedures for mounting loose-part sensors and a method for locating loose parts that is based on amplitude (rather than time-of-arrival) differences.

BOILING WATER REACTOR STABILITY MONITORING

There is currently a large international effort to better understand the nature of neutron noise in boiling-water reactors (BWRs). Potential applications of noise analysis for safety assessment in BWRs include on-line measurement of void fraction,^{5,6} detection of in-core component vibration and impacting,⁷ and monitoring of core power stability. We previously assessed void and vibration monitoring, and now we are evaluating the use of neutron noise for continuous on-line stability monitoring. Results indicate that changes in stability parameters (such as the decay ratio) can be detected by performing time series analysis of neutron noise from power range monitor signals in BWRs.⁸ We have recommended that NRC perform an in-plant demonstration of such a continuous stability monitor to further assess the practicability of this methodology.

References

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2. F. J. Sweeney, "Modeling the Local Component of In-Core Neutron Detector Noise in a BWR," *Trans. Amer. Nucl. Soc.* (to be published, November 1979).
3. F. J. Sweeney, "Sensitivity of Detecting BWR Control Rod Vibrations Using Neutron Noise," *Trans. Amer. Nucl. Soc.* (to be published, November 1979).
4. R. C. Kryter and C. W. Ricker, *Characteristics and Performance Experience of Loose-Part Monitoring Systems in U. S. Commercial Power Reactors*, NUREG/CR-0524 (December 1978).
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7. D. N. Fry, R. C. Kryter, M. V. Mathis, J. E. Mott, and J. C. Robinson, "Use of Neutron Noise to Detect BWR-4 In-Core Instrument Tube Vibrations and Impacting," *Nucl. Technol.* 43, 42-54 (April 1979).
8. B. R. Upadhyaya and M. Kitamura, "Monitoring BWR Stability Using Time Series Analysis of Neutron Noise," *Trans. Amer. Nucl. Soc.* (to be published, November 1979).

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D. N. FRY -- DEVELOPMENT GROUP

I&C DIVISION -- ORNL

NOISE ANALYSIS: A USEFUL TOOL TO DIAGNOSE AND
ASSESS SAFETY-RELATED PROBLEMS IN NUCLEAR PLANTS

SEVENTH WATER REACTOR SAFETY RESEARCH
INFORMATION MEETING, NATIONAL BUREAU OF STANDARDS

NOVEMBER 9, 1979

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WE HAVE RECENTLY USED NOISE ANALYSIS METHODS TO
AID NRC IN

1. ASSESSMENT OF EX-CORE NUCLEAR SIGNAL
FLUCTUATIONS AT FORT ST. VRAIN

AND

2. POST-ACCIDENT ASSESSMENT OF REACTOR VESSEL
INTERNALS AND OPERATIONS AT THREE MILE
ISLAND 2

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WE HAVE ALSO CONTINUED TO:

- * ACQUIRE BASELINE NEUTRON NOISE SIGNATURES FROM OPERATING PLANTS
- * DEVELOP ANALYTICAL TECHNIQUES TO PREDICT THE SENSITIVITY OF NOISE ANALYSIS FOR DETECTING POSTULATED ANOMALIES
- * CONDUCT EXPERIMENTS TO EVALUATE THE METHODS CURRENTLY USED BY THE NUCLEAR INDUSTRY TO DETECT, LOCATE, AND DETERMINE THE SIZE OF LOOSE PARTS IN THE PRIMARY COOLANT SYSTEM
- * DETERMINE THE FEASIBILITY OF USING NEUTRON NOISE FOR CONTINUOUS, ON-LINE MONITORING OF STABILITY IN BWRs

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WE PROVIDED ASSISTANCE TO NRC IN ASSESSING
A PROBLEM AT FORT ST. VRAIN

THE PROBLEM:

AT TIMES DURING NORMAL OPERATION, LARGE OFFSETS
OCCUR IN EX-CORE NEUTRON FLUX CHANNELS

THESE OFFSETS ARE PRESUMED TO BE CAUSED BY
GRAPHITE BLOCK MOVEMENTS IN THE CORE

OUR ASSISTANCE:

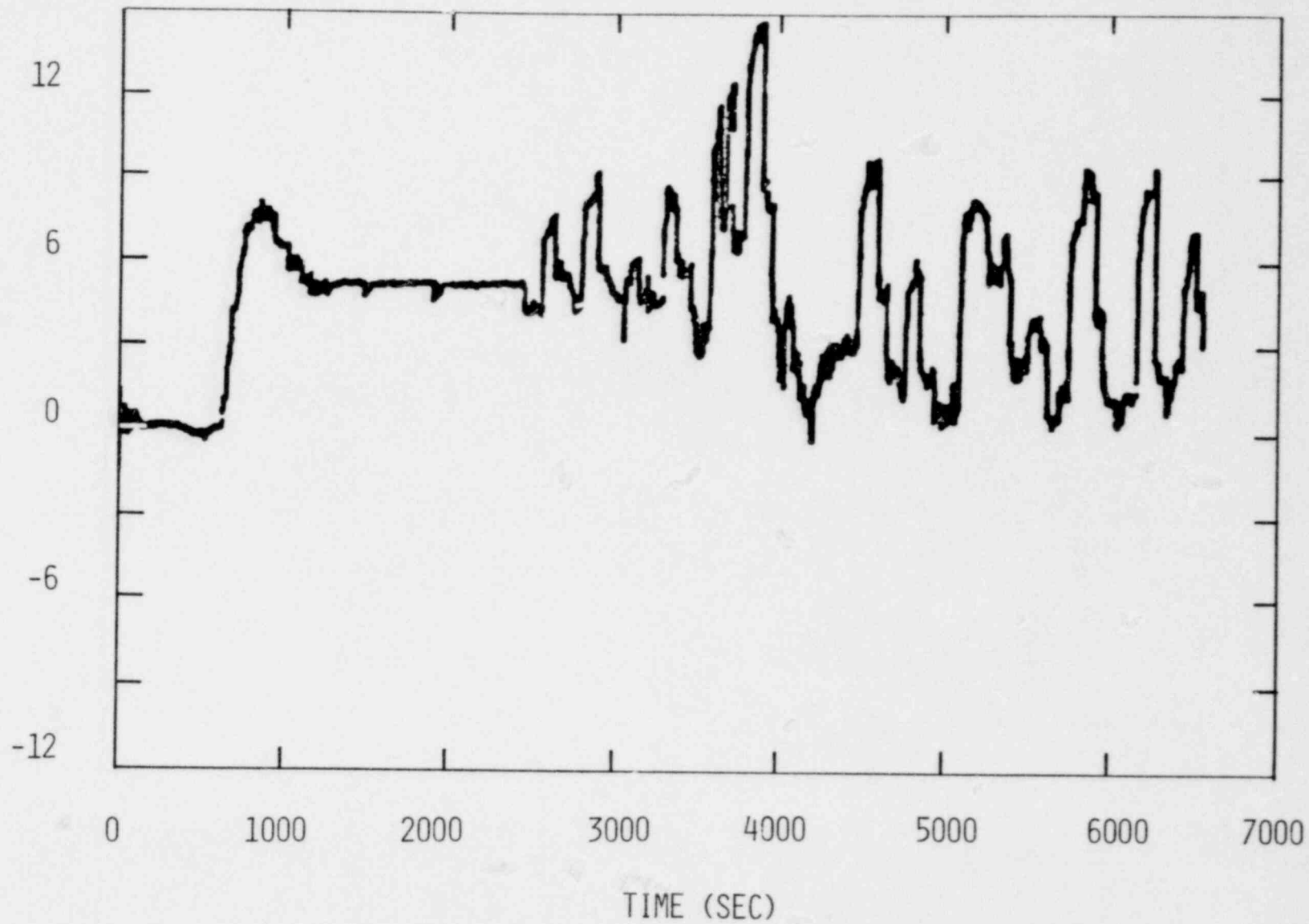
WE ANALYZED SIGNALS FROM EX-CORE NEUTRON DETECTORS
AND PRESSURE VESSEL DISPLACEMENT PROBES TO INFER
THE PREDOMINANT DIRECTION OF MOVEMENT

1607 138

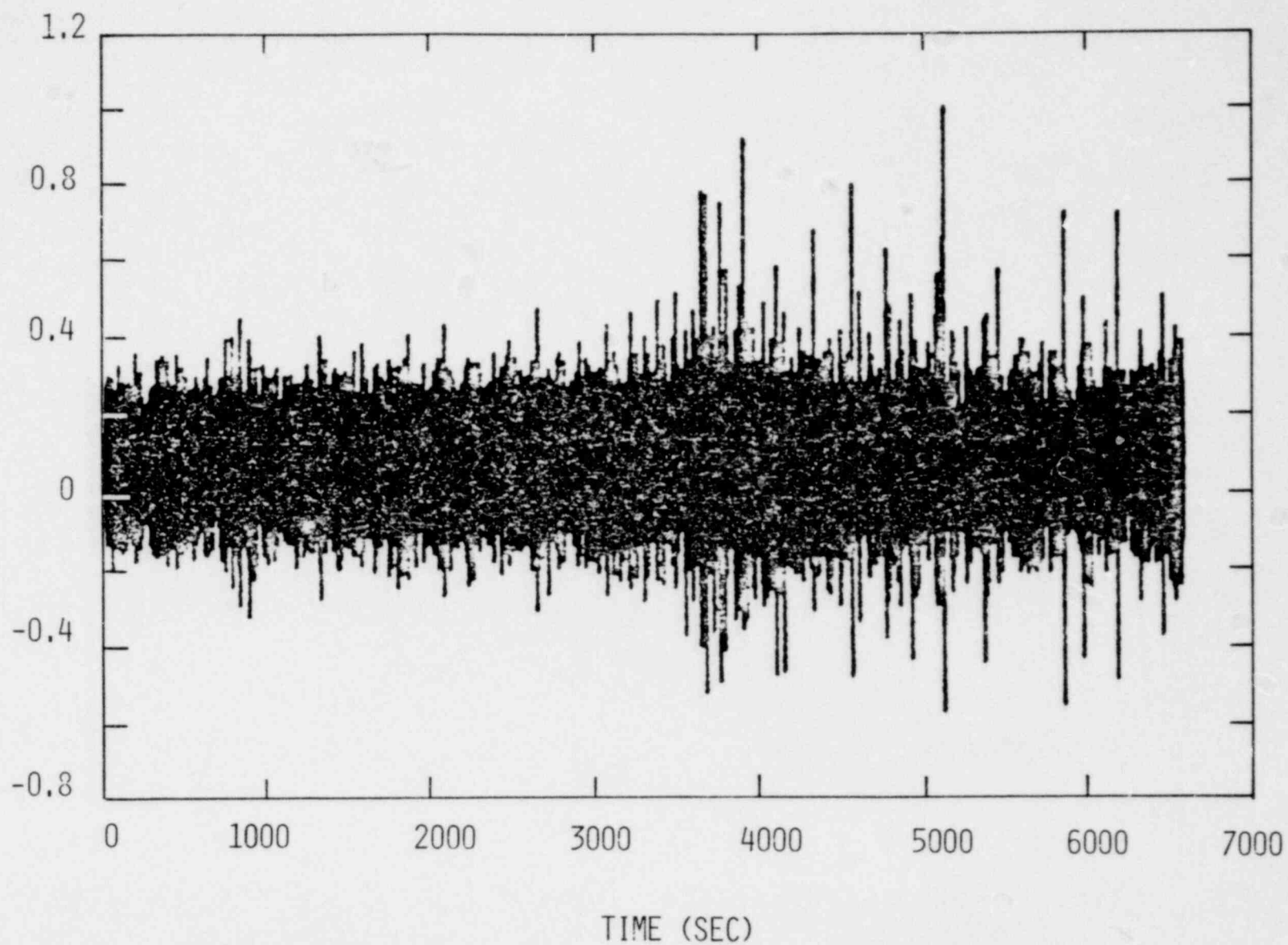
INCREASED ACTIVITY ON DISPLACED PROBES CAN BE ASSOCIATED
WITH OFFSETS IN NUCLEAR SIGNALS

EXCORE
NEUTRON
SIGNAL
(% FULL POWER)

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PRESSURE
VESSEL
DISPLACEMENT
(MILS)



1607 140

THEREFORE, NOISE ANALYSIS AIDED NRC's
EVALUATION OF FORT ST. VRAIN OSCILLATIONS

1. COINCIDENCE ANALYSIS OF NEUTRON FLUX AND PCRV
DISPLACEMENT SIGNALS YIELDED INFORMATION ABOUT
THE DIRECTION OF CORE MOTION
2. DATA OBTAINED SO FAR SUGGEST THAT CORE MOVES
PREFERENTIALLY IN A NW-SE DIRECTION

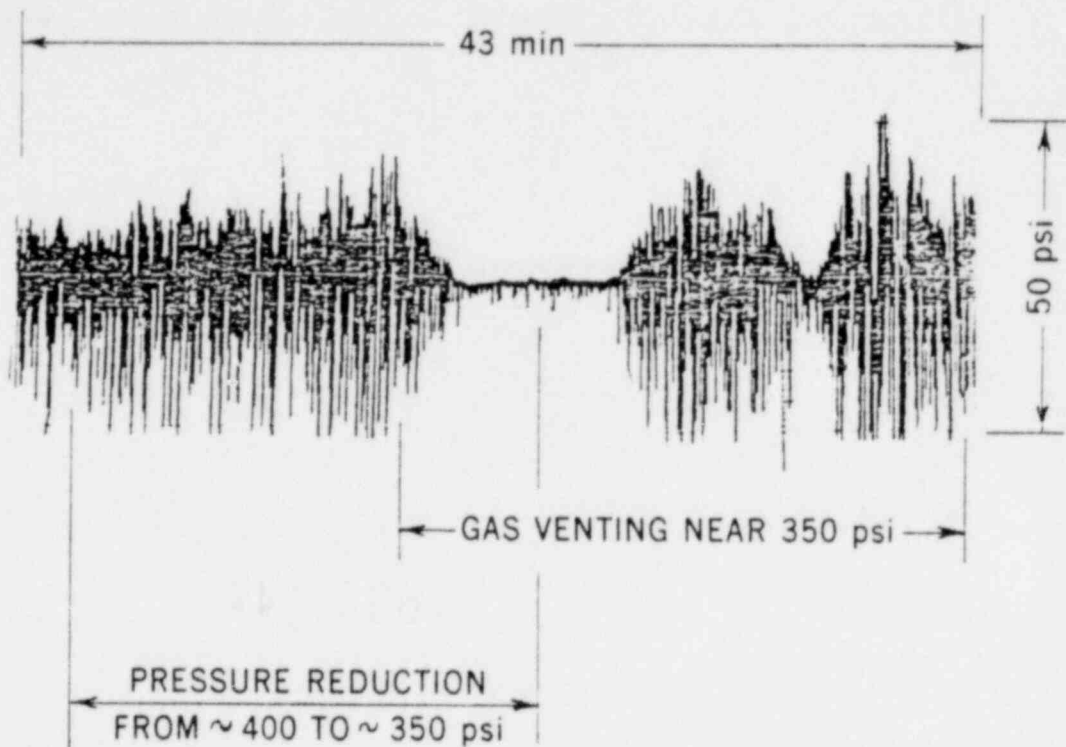
1607 141

WE PERFORMED ON-SITE DIAGNOSTIC
MEASUREMENTS AT TMI TO:

- * VERIFY THE POST-ACCIDENT STRUCTURAL INTEGRITY
OF THE REACTOR VESSEL INTERNALS
- * VERIFY THAT A NONCONDENSIBLE GAS BUBBLE IN THE
PRIMARY SYSTEM DID NOT POSE A THREAT TO CORE
COOLING
- * VERIFY THAT BULK COOLANT BOILING WAS NOT OCCURRING
IN THE CORE DURING ONE-PUMP AND NATURAL CONVECTION
COOLING
- * PROVIDE GUIDANCE TO PLANT OPERATORS DURING DEGASIFI-
CATION OF THE PRIMARY SYSTEM

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A PORTION OF STRIP CHART RECORDING SHOWS HOW NOISE
ANALYSIS MADE ITS MOST VALUABLE AND DIRECT
CONTRIBUTION DURING PRIMARY COOLANT DEGASIFICATION OPERATIONS



1607 143

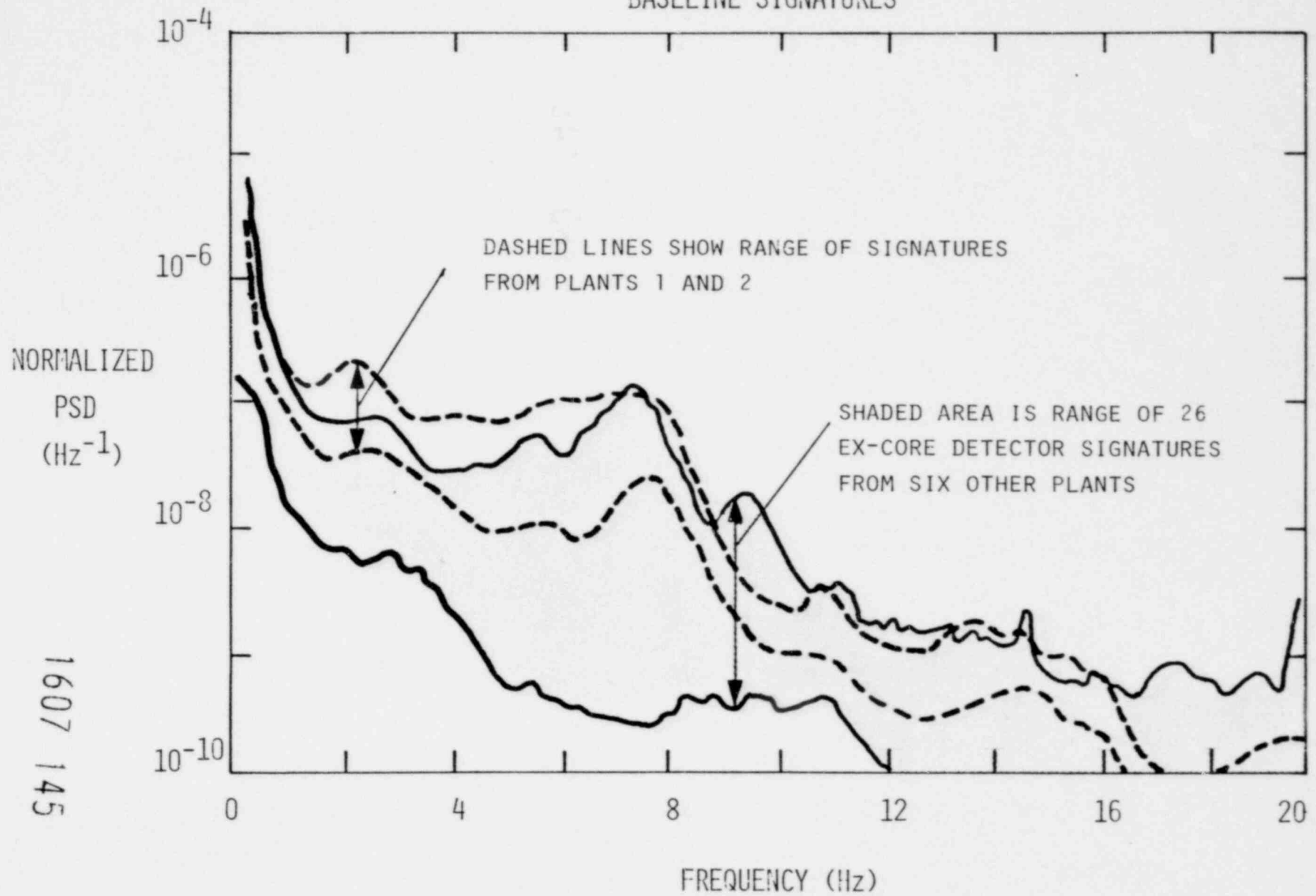
WE OBTAINED BASELINE NEUTRON NOISE SIGNATURES
FROM EX-CORE DETECTORS IN TWO PWRs

TO BE USED TO ASSESS

FUTURE CORE OR CORE BARREL VIBRATION
PROBLEMS IF THEY OCCUR

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BOTH PLANT SIGNATURES ARE CONSISTENT WITH PWR
BASELINE SIGNATURES



WE ARE DEVELOPING ANALYTICAL METHODS
TO UNDERSTAND THE BASIC PROCESSES
WHICH PRODUCE REACTOR NOISE
AND
TO ESTABLISH THE RANGE AND DETECTABILITY
OF POSTULATED ANOMALIES

1607 146

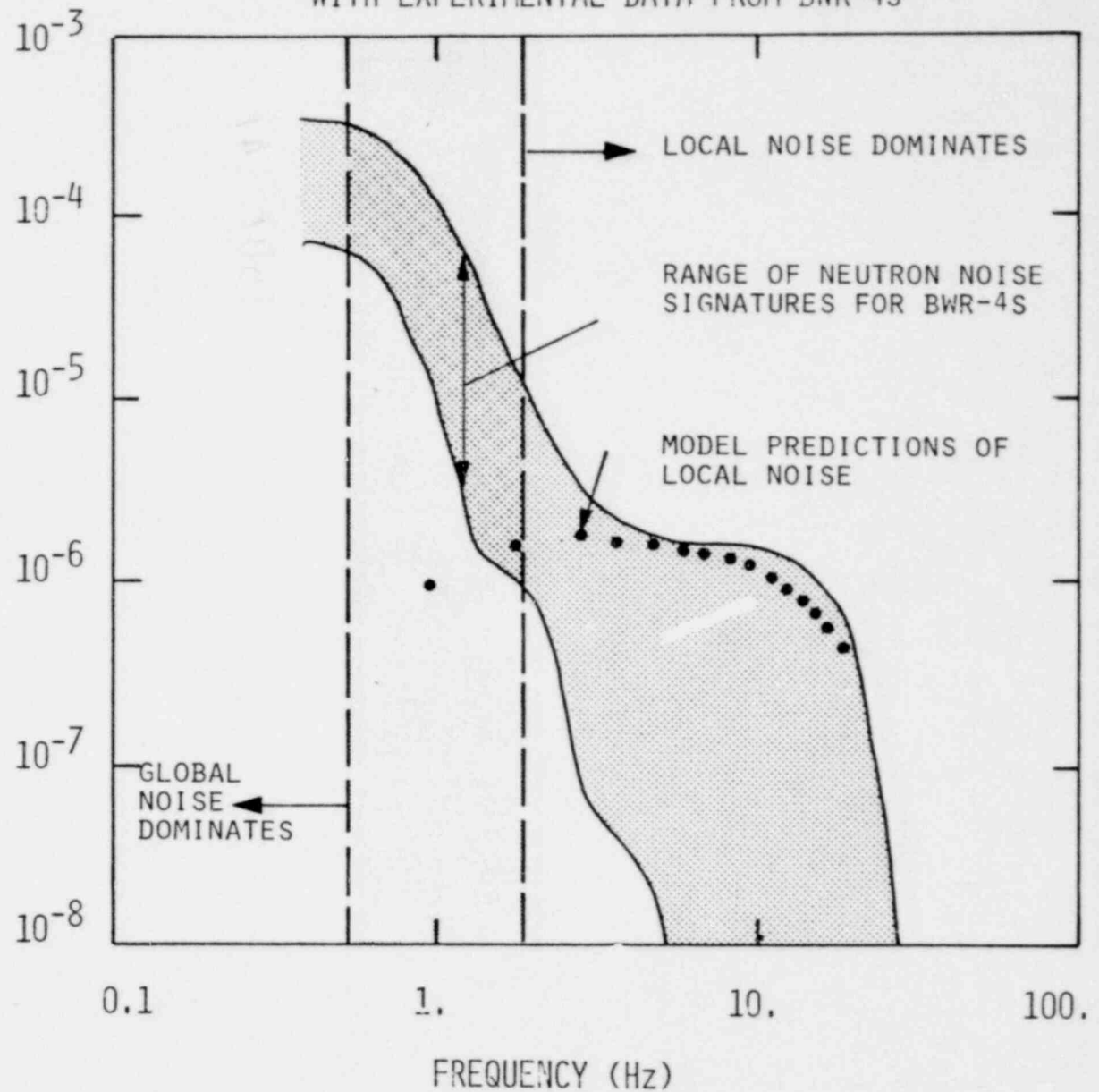
THE METHODOLOGY WAS EVALUATED BY:
COMPARING CALCULATED BWR BACKGROUND NEUTRON NOISE
CAUSED BY BOILING WITH SIGNATURES OBTAINED FROM FIVE BWR-4s
AND
BY ESTIMATING THE MINIMUM AMOUNT
OF CONTROL ROD VIBRATION THAT CAN BE
RELIABLY DETECTED IN A BWR-4 IN THE PRESENCE OF THE
NORMAL BOILING NOISE

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NORMALIZED
CPSD FOR
C&D DETECTORS
IN A BWR-4
(Hz⁻¹)

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PRELIMINARY MODEL PREDICTIONS COMPARE WELL
WITH EXPERIMENTAL DATA FROM BWR-4s



WE ARE NOW APPLYING THIS METHODOLOGY
TO CALCULATE NEUTRON NOISE CAUSED BY POSTULATED

* VIBRATION OF CORE COMPONENTS IN PWRs

AND

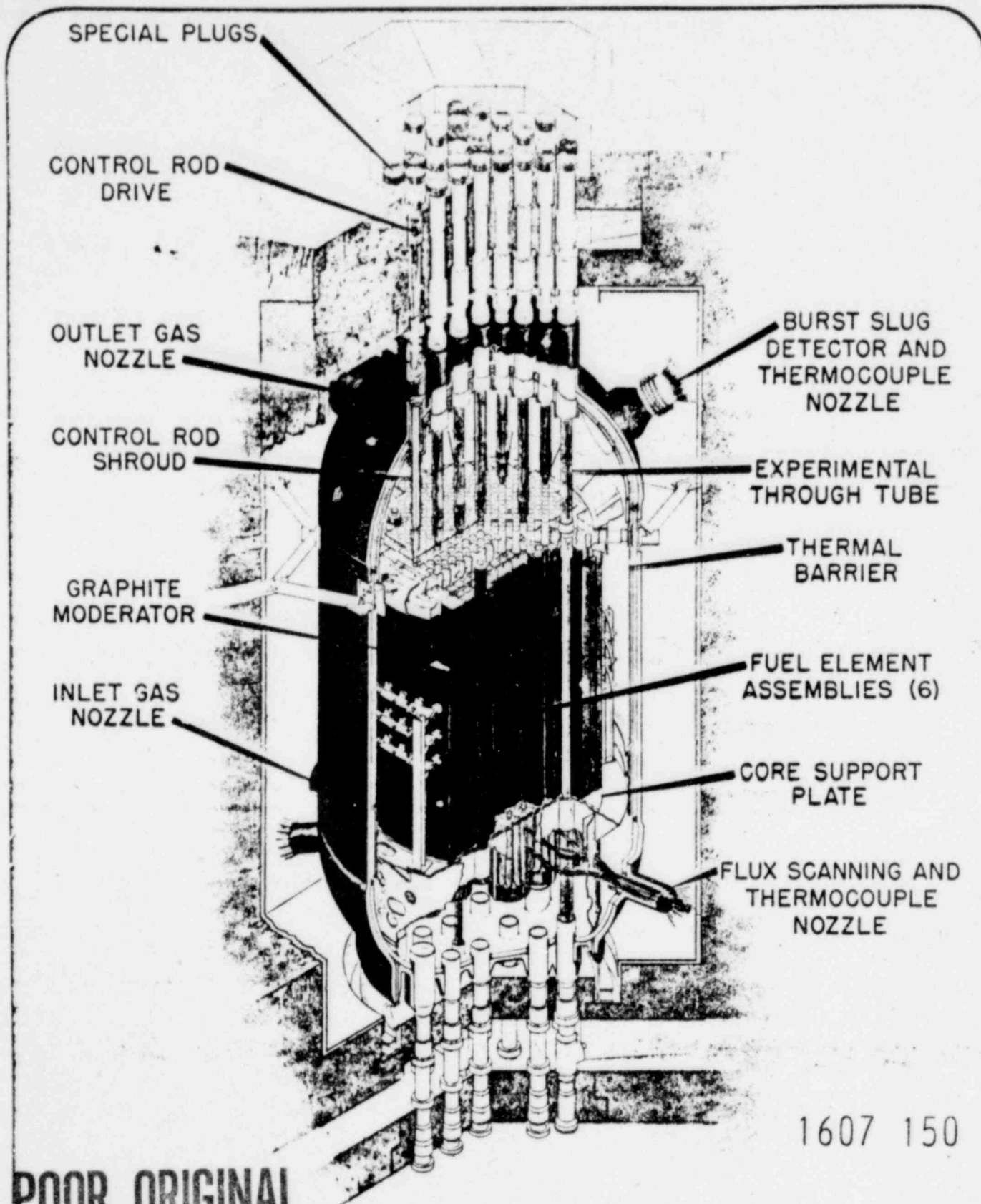
* CORE VOIDING IN PWRs

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8-11-1981

OCT 1981

WE EVALUATED LOOSE PARTS LOCATION METHODS BY LOCATING
FOUR ACCELEROMETERS ON THE LOWER HEAD OF THE EGCR AND
THEN IMPACTING THE VESSEL BETWEEN TWO OF THE ACCELEROMETERS.



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POOR ORIGINAL

EXPERIMENTAL GAS-COOLED REACTOR

TO INFER THE LOCATION OF IMPACTS, WE HAVE
EXPLORED THE USE OF ENERGY ATTENUATION
WITH IMPACT-TO-SENSOR DISTANCE

THE METHOD APPEARS TO GIVE GOOD RESULTS

BUT

IT DOES REQUIRE DETAILED KNOWLEDGE
OF THE SOUND TRANSMISSION PATH,
WHICH IS NOT EASILY OBTAINED FOR COMPLEX STRUCTURES

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IN THE QUIET ENVIRONMENT OF THE EGCR WE CAN LOCATE
THE IMPACT SITE TO WITHIN 12 INCHES IF WE ASSUME
THAT THE IMPACT IS ON THE VESSEL WALL

(VESSEL DIAMETER IS ~ 20 FT)

<u>IMPACT NO.</u>	<u>ERROR IN LOCATION (IN.)</u>
1	8.1
2	10.1
3	10.4
4	11.7
5	6.9
6	16.3
7	12.0
8	12.2
9	11.3
10	9.6
11	12.7
12	12.7

MEAN ERROR = 11.2 INCHES

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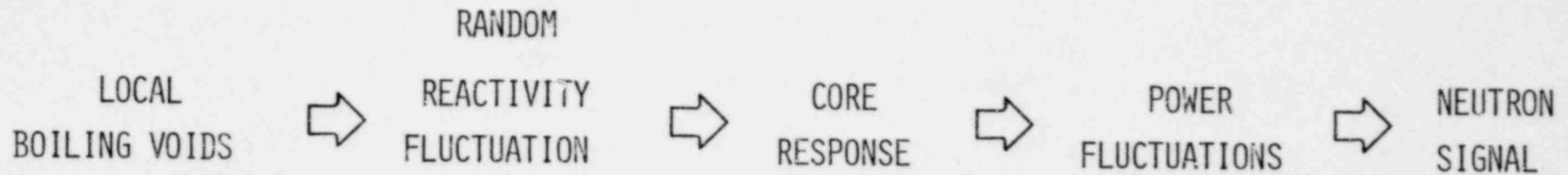
CHANGES IN CORE DESIGN OR OPERATING
STRATEGY MAY AFFECT STABILITY MARGIN IN BWRs

THEREFORE

- * THE PURPOSE OF OUR PROGRAM IS TO DETERMINE THE FEASIBILITY OF USING NOISE SIGNALS TO DETECT CHANGES IN CORE STABILITY WITHOUT DISTURBING PLANT OPERATION
- * OUR METHOD OBTAINS DECAY RATIOS OR OTHER SIMILAR STABILITY PARAMETERS FROM NEUTRON NOISE SIGNALS USING TIME SERIES ANALYSIS
- * WE HAVE DETERMINED THE FEASIBILITY OF OUR METHOD BY USING EXISTING NEUTRON SIGNALS RECORDED AT AN 1100 MWe BWR/4 AT FULL POWER

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NORMAL BOILING CAUSES RANDOM REACTIVITY FLUCTUATION WHICH STIMULATES THE CORE



FROM THIS

WE CAN INFER CHANGES IN THE DYNAMIC RESPONSE OF

THE CORE BY MONITORING THE NEUTRON NOISE

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UNIVARIATE AND MULTIVARIATE TIME SERIES ANALYSIS
OF NEUTRON SIGNALS WAS USED TO INFER CORE STABILITY

UNIVARIATE DECAY RATIO = 0.024

MULTIVARIATE DECAY RATIO = 0.020

THEREFORE

WE RECOMMEND THAT THE UNIVARIATE METHOD

BE USED TO MONITOR STABILITY

BECAUSE

IT IS EASIER TO IMPLEMENT AND SEEMS TO YIELD

A CONSERVATIVE ESTIMATE OF STABILITY

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FROM THESE STUDIES AND EXPERIENCES,

WE CONCLUDE THAT NOISE ANALYSIS HAS A PLACE
AMONG OTHER METHODS IN DIAGNOSING AND ASSESSING
SAFETY-RELATED PROBLEMS IN NUCLEAR PLANTS

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